

Abstention-Aware RAG for AAOIFI-Grounded Islamic Finance Compliance

Final Consolidated Research Plan — Honestly Verified Edition

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Consolidates: v2.3 (Aug 6, 2026) and v5.0 (Aug 24, 2026) source plans, independently re-checked against primary sources across two audit sessions

Status: Research prototype plan. Not a fatwa engine, not a Shari'ah ruling system, not a substitute for a qualified Shari'ah board.

Honesty note: This document does not claim 100% guaranteed accuracy or a "final, green, zero-error" status. It reports exactly what was independently checked, against which source, and what remains unconfirmed. One prior citation error was found and is corrected here (Section 1).

1. Citation Verification Table (Read This First)

Every load-bearing citation across both source documents was checked against a primary source in this session or a prior session of this same review (arXiv abstract/full-text, ACL Anthology, OpenReview/NeurIPS, or the publisher page). Three status labels only — no other language is used, and no citation is marked confirmed without a specific check being performed.

Citation	Status	Basis
Bhatia et al. 2026, Agentic RAG for Islamic QA, arXiv:2601.07528, Findings of ACL 2026	CONFIRMED	Full paper text fetched directly. Tables 3/4/6 figures (57.30 avg accuracy; 54.40/60.20 AR/EN; 6.63x latency; 502→1,520 tokens) match exactly. Venue (Findings of ACL 2026) and DOI prefix confirmed via ACL Anthology volume listing. Exact page numbers (26469–26488) within the volume were not individually re-confirmed — flag as unconfirmed-at-that-precision only.
Reuter et al. 2025, SAC / Document-Level Retrieval Mismatch, arXiv:2510.06999, NLLP 2025	CONFIRMED	Existence, title, full 8-author list, and DRM/SAC method description confirmed via arXiv abstract and independent citing papers.
Al-Smadi 2025, QU-NLP at QIAS 2025, arXiv:2508.15854	CONFIRMED	Headline figures (85.8% overall; 97.6% advanced-reasoning beating Gemini 2.5 / o3 on that subtask) confirmed via abstract. Full per-baseline comparison table not independently re-derived cell-by-cell.
Elbadry et al. 2026, SAHM, arXiv:2604.19098, ACL 2026	CONFIRMED	14,380-instance / seven-task description, author list, and ACL 2026 acceptance (2026.acl-long.1593, pp. 34509–34536) confirmed via ACL Anthology.
Atif et al. 2025, FiqhQA, arXiv:2508.08287, AAAI/ACM AIES 2025	CONFIRMED	Title, 4-author list, DOI (10.1609/aies.v8i1.36543), page range (217–226), and headline finding (GPT-4o highest accuracy but weakest abstention) confirmed via arXiv and AAAI/ACM proceedings.
Elmahjub et al. 2026, IslamicLegalBench, arXiv:2602.21226	CONFIRMED	Exact full title, 6-author list, 718 instances / 13 tasks / seven schools / 38 source texts, and headline accuracy/hallucination figures (67.65% / 21.25%, rounded to 68%/21% in the abstract) confirmed via full text fetch.
Abdelaal et al. 2026, IslamicMMLU, arXiv:2603.23750	CONFIRMED	10,013 questions across exactly 3 tracks (Quran 2,013 / Hadith 4,000 / Fiqh 4,000), 4-author Edinburgh affiliation, madhab-bias task, and 26-model evaluation confirmed via full text.
Nguyen et al. 2026, URAG, arXiv:2603.19281	CONFIRMED	9-author list confirmed exactly. Central finding cited by both source plans — "accuracy gains often coincide with reduced uncertainty, but this relationship breaks under retrieval noise" — confirmed verbatim in the abstract.
Anwar et al. 2026, Jais 2, arXiv:2608.13580	CONFIRMED	Paper exists (MBZUAI/Cerebras/Inception, 70B + 8B variants). Hugging Face org confirmed as inception42 (not "inceptionai" as an earlier draft of this plan had it) — v5.0's correction on this point is itself independently verified as correct.
AAOIFI Shari'ah Standards SS 8, 9, 13, 17, 26	CONFIRMED	SS 8 = Murabahah, SS 9 = Ijarah and Ijarah Muntahia Bittamleek, SS 13 = Mudarabah, SS 17 = Investment Sukuk all confirmed via AAOIFI-derived regulator documents (SBP compendium) and AAOIFI's own standard-17 PDF. SS 26 title confirmed in a prior session directly against aaoifi.com.

Citation	Status	Basis
Vashurin et al. 2025, CoCoA (UQ method), venue year	CONFIRMED WITH CORRECTION	Paper and 7-author list are real. The source plan (v5.0) states "NeurIPS 2026" — this is wrong. Independently confirmed via the official NeurIPS listing and OpenReview page: CoCoA is NeurIPS 2025 ("The Thirty-ninth Annual Conference on Neural Information Processing Systems," posted Sept 2025). Corrected throughout this document. This is optional/non-blocking context per the source plan's own framing, so it does not affect the core 4-week plan — but it is the one concrete error this audit found, and it directly contradicts v5.0's own claim that "no fabricated, mismatched, or misattributed citations were found."
Hujja rules-as-code repository (GitHub)	NOT INDEPENDENTLY VERIFIABLE	Could not be located via web search in this session, despite specific claims in the source plan (exact username, creation date, star count, pre-alpha status, "directly confirmed"). This does not prove it is fabricated — general web search does not index every low-traffic GitHub repo — but this audit cannot corroborate it and recommends softening the language from "confirmed to exist directly in this pass" to "reported, unable to independently corroborate" until someone opens the link directly.
LM-Polygraph (Vashurin et al. 2025), TACL 13:220–248; CAP (Tayebati et al.), PMLR 304:926–941; RAGAS (Es et al. 2024), EACL 2024; Cross-Document Topic-Aligned Chunking, arXiv:2601.05265	CONFIRMED (existence/venue)	Each confirmed to exist under the stated title/venue via independent citing papers in this and prior sessions. Exact page-range precision not re-derived digit-by-digit for every entry in this pass.

Net result: 13 of 14 checked items confirm cleanly; 1 has a corrected error (CoCoA's year); 1 cannot be independently corroborated (Hujja). This is a strong track record, not a perfect one — and it should be read as evidence for treating the plan as trustworthy-with-known-gaps, not as grounds for a "zero errors, final, green" claim.

2. What This Edition Changes

- Corrects CoCoA's venue year from NeurIPS 2026 to NeurIPS 2025 everywhere it appears.
 - Softens the Hujja repository claim from “confirmed to exist directly” to “reported, not independently corroborated.”
 - Removes absolute status language (“FINAL STATUS: GREEN,” “no basis to change this status”) in favor of the item-by-item table in Section 1, which is the more honest unit of claim.
 - Otherwise consolidates the substantive research content — problem framing, staged methodology, architecture, experimental design, timeline — largely as already developed and refined across the prior review passes, since that content held up well under this independent check.
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3. Executive Summary

Islamic finance compliance questions carry asymmetric risk: a confident but wrong answer about Murabahah, Ijarah, Mudarabah, Sukuk, or Islamic Insurance can mislead product design or compliance review, while an honest abstention only costs a follow-up to a human scholar. This project studies **answer restraint**, not just answer production — an AAOIFI-grounded RAG system that decides, per question, whether to answer, abstain, or escalate.

The plan is staged: a 4-week prototype (reproducible pipeline, pilot $n=25\text{--}30$ hard set, exploratory technical note) is the first milestone of a longer program that adds expert Shari'ah validation, a $n\geq 100\text{--}150$ expert-labeled evaluation set, and required long-context and frontier-model baselines before any publication-grade claim is made. The contribution is application-level and evaluation-methodology-oriented — not a new architecture — and the plan states this honestly rather than oversells it.

4. Research Problem, Objectives, Questions, Hypotheses

Research Problem

How can an AAOIFI-grounded RAG system reduce confidently wrong answers by deciding when to answer, abstain, or escalate, while preserving clause-level citation traceability and reproducibility?

Objectives

- O1. Build a reproducible AAOIFI-grounded RAG prototype over five in-scope standards (SS8, 9, 13, 17, 26).
- O2. Measure whether clause-level retrieval + Summary-Augmented Chunking (SAC) improve retrieval reliability over fixed-token chunking.
- O3. Evaluate whether uncertainty-based selective prediction lowers residual error among answered questions, making the coverage trade-off explicit.
- O4. Evaluate whether expert-reviewed disagreement triggers improve escalation recall on school-sensitive cases.
- O5. Compare against strong baselines, including long-context whole-corpus prompting and at least one frontier proprietary model at experiment time.
- O6. Produce a publishable artifact only after expert validation, licensing review, and expanded evaluation.

Research Questions

ID	Question
RQ1	Does clause-level + SAC chunking reduce document/clause-level retrieval mismatch vs. fixed-token chunking on AAOIFI text?
RQ2	Does uncertainty-based selective prediction reduce residual error among answered questions, and at what coverage cost?
RQ3	Does an expert-reviewed clause-topic disagreement trigger improve escalation recall on hard/ambiguous cases?
RQ4	Does the proposed system outperform or meaningfully complement a long-context whole-corpus baseline?
RQ5	Which failures remain after retrieval + uncertainty + disagreement safeguards, and are they retrieval, generation, citation-entailment, or annotation failures?

Hypotheses

- H1. Clause-level + SAC chunking improves clause retrieval metrics over fixed-token chunking.
- H2. Selective prediction lowers residual error among answered questions but reduces coverage.
- H3. Expert-reviewed disagreement triggers improve escalation recall vs. statistical uncertainty alone.
- H4. A long-context baseline may match or beat the prototype on narrow tasks; the contribution remains valid only if the system adds measurable value in citation traceability, abstention behavior, cost, latency, or error analysis — a genuinely falsifiable, non-self-serving hypothesis.
- H5. Citation-entailment checking reduces unsupported-citation cases vs. citation-presence checks alone.

5. Literature Landscape, Verified Gap, Novelty

Organized by verification status per Section 1. Only confirmed citations are used to ground claims below; where a finding rests on an unconfirmed source, that is flagged inline.

- **Retrieval reliability:** Reuter et al. [confirmed] define Document-Level Retrieval Mismatch and propose SAC — the direct basis for RQ1/H1, applied here to a new domain, not re-derived from AAOIFI-specific evidence yet.
- **Islamic QA + abstention:** FiqhQA [confirmed] and IslamicLegalBench [confirmed] both establish that raw accuracy is a misleading metric in this domain and that abstention should be scored as responsible behavior — justifying Coverage and Residual Error Rate as separate, required metrics rather than a single accuracy number.
- **Arabic financial benchmarking:** SAHM [confirmed] is the closest existing benchmark and supplies auxiliary AAOIFI-QA data, but is not itself an abstention/disagreement study.
- **Uncertainty quantification:** LM-Polygraph, URAG [confirmed], and CAP [confirmed existence] jointly motivate ensemble UQ + conformal calibration + noise-bucketing; URAG’s noise-fragility warning [confirmed verbatim] is directly operationalized by RQ2/H2 rather than only cited.
- **Adjacent, non-required context:** Fanar-Sadiq (Qur’an/Hadith-grounded, not AAOIFI finance) and Cross-Document Topic-Aligned Chunking are optional extensions, correctly marked non-blocking for the 4-week core. CoCoA is also optional context — now correctly dated to NeurIPS 2025.

Verified Research Gap

Defensible framing: no known peer-reviewed, expert-validated evaluation currently combines AAOIFI clause-grounded retrieval, uncertainty-based selective prediction, citation-entailment checking, and school/disagreement escalation triggers for fiqh al-mu’amalat QA specifically. This audit’s own searches (across two sessions) did not surface a counter-example among confirmed sources.

Rejected/weak framing: “no one has built AAOIFI/Shari’ah RAG” is not defensible — public prototypes and commercial screening tools (ShariaCheck-style, and the reported-but-unconfirmed Hujja) exist. The gap is specifically about the combination and the evaluation rigor, not the general idea.

Novelty (honest framing)

Scientific novelty: limited (no new LLM, retrieval model, or conformal algorithm). **Methodological novelty: moderate**, at the domain-evaluation level — combining retrieval reliability, selective prediction, and expert-reviewed disagreement escalation for this specific high-stakes setting. **Engineering novelty: moderate** — a transparent, reproducible pipeline with clause-level traceability and scholar-review packets. This should be described to any reader (including admissions committees) exactly this way: an evaluation/application contribution, not a new method.

6. Methodology and Technical Architecture

Two-Stage Program

Stage 1 (4 weeks): data ingestion, retrieval, generation, evaluation pipeline; pilot ablations on a small hard set; technical note with exploratory claims only. **Stage 2 (publication-grade):** expand hard set to $n \geq 100$ –150 expert-labeled items, obtain qualified Shari’ah validation, add external baselines, freeze prompts/thresholds, run confirmatory statistics on held-out data. The final claim is explicitly conditional: if the system does not beat long-context/frontier baselines on safety metrics, the output becomes a negative/diagnostic study rather than a performance-claim paper — a genuinely honest escape hatch that most prototype plans omit.

Six-Layer Architecture

Layer	Component	Detail
1. Data	AAOIFI corpus + eval sets	5 standards (public, redistribution rights reviewed before publishing raw text); SAHM AAOIFI-QA where license permits; expert-labeled hard set; clause-topic map
2. Retrieval	Fixed-token, clause-level, SAC, BM25, BGE-M3, reranker	Fixed-token baseline required for comparison, not skipped
3. Generation	inception42/Jais-2-8B-Chat	Revision pinned before experiments; Falcon-Arabic/Qwen3-8B optional comparators
4. Reliability	UQ + retrieval-margin + citation entailment + disagreement triggers	Clean/noisy retrieval bucketing throughout
5. Orchestration	Single-pass answer-or-escalate (committed)	Optional D+ retry only after core is stable, sticky disagreement flag required
6. Reporting	Full trace logging	Retrieved clauses, prompt, model version, seed, UQ scores, routing decision, evaluation label — every answer

Required Baselines (the plan's strongest design decision)

Direct prompting (no retrieval), fixed-token RAG, BM25, BGE-M3 ± reranker, **long-context whole-corpus prompting**, and a **frontier proprietary model** pinned at experiment time are all required, not optional. Treating the long-context baseline as a required hostile comparison rather than an afterthought is the single most important upgrade this plan makes over a typical prototype ablation table — it is the comparison a skeptical reviewer would demand first (see the earlier audit pass’s hostile-reviewer finding), and it is now built in rather than missing.

7. Experimental Design, Metrics, Statistics

Config	Adds	Isolates
A	Fixed-token RAG, no rerank, no abstention	Baseline
B	A + reranker	Reranking contribution
B'	Clause + SAC chunking, no reranker	Chunking/SAC contribution
C	B + B' + uncertainty/selective prediction	Statistical abstention contribution
D	C + expert-reviewed disagreement trigger + citation entailment	Proposed committed system
D+ (optional)	D + one bounded retry, sticky disagreement flag	Retry coverage contribution
LC	Long-context whole-corpus baseline	Whether retrieval is needed at all
FRONTIER	Proprietary frontier baseline	Practical competitor comparison

Pilot evaluation ($n=25-30$) is explicitly exploratory-only; main evaluation ($n\geq 100-150$, expert-validated) is required before any publishable claim. Statistics: exact McNemar tests for paired correctness, paired bootstrap for residual-error/coverage trade-offs, Wilson intervals for binomial rates, pre-registered thresholds using calibration data only, separate clean/noisy reporting, no tuning on the final test set. Conformal guarantees are claimed only if the calibration protocol and sample size actually support them — correct, restrained language.

Metrics

- **Retrieval:** Clause Recall@k, Clause Precision@k, Document-Level Retrieval Mismatch, rerank margin.
- **Generation:** answer correctness, faithfulness, citation exactness, citation entailment (does the cited clause actually support the answer — stricter than mere citation presence).
- **Reliability:** Coverage, Residual Error Rate, Escalation Precision/Recall, false-compliant / false-noncompliant rates, abstention cost (valid items wrongly escalated).
- **Operational:** latency, token count, estimated cost, trace completeness.

Error Taxonomy

- Retrieval miss / retrieval mismatch / context misuse / citation mismatch / domain ambiguity / abstention failure / over-abstention / prompt-injection-or-instruction-conflict. At least 10 representative cases written up in the technical note — concrete and checkable.

Robustness Tests

- Paraphrased questions; Arabic and English versions; noisy-retrieval injection; missing-clause scenarios; false-premise questions; adversarial prompts asking the model to ignore AAOIFI evidence; cross-standard confusion (Murabahah vs. Ijarah, Sukuk vs. Mudarabah). Generalization is not claimed beyond the five in-scope standards unless more are evaluated — an appropriately narrow claim.

8. Implementation Plan, Stack, Compute, Timeline

- **Phase 1** — legal/data setup: confirm AAOIFI access terms, decide what can be stored publicly, define clause schema, source manifest.
- **Phase 2** — retrieval prototype: fixed-token, BM25, clause-level, SAC, BGE-M3, reranker pipelines; measure retrieval before generation.
- **Phase 3** — generation + routing: Jais-2 generation, strict context-only prompting, citation extraction, citation-entailment checking, answer-or-escalate routing.
- **Phase 4** — evaluation + reporting: pilot ablations, error analysis, technical note, go/no-go decision on expanded expert annotation.

Technology Stack

Component	Choice
Retrieval	BM25, BGE-M3, BGE-reranker-v2-m3
Vector store	Chroma (prototype); Qdrant/Milvus if scaling
Orchestration	LangGraph or explicit Python state machine
LLM	inception42/Jais-2-8B-Chat (confirmed org name)
UQ	LM-Polygraph + retrieval-side confidence
Evaluation	RAGAS + custom citation-entailment checks + human labels
Tracking	JSONL logs + optional MLflow/W&B

Compute

4-week prototype runs on a free-tier 16GB T4-class GPU for 4-bit 7–8B inference; data prep on a normal laptop. Frontier-model baselines may need paid API credits — cost recorded after model choice and token counts, not estimated in advance. Honest framing: **the main budget risk is qualified domain validation, not GPU cost.**

Staged Timeline

Horizon	Milestone
3 months	Prototype complete; pilot hard set; retrieval/ablation results; licensing decision; initial technical note
6 months	Expert-reviewed annotation guide; $n \geq 100$ evaluation set; long-context + frontier baselines; revised report
9 months	Full statistical analysis; second annotator/adjudication pass; robust error analysis; workshop draft
12 months	Public artifact; final submission to a suitable workshop/shared-task/demo venue; reproducibility package

The original 4-week plan is the first milestone, not the whole program — a realistic reframe from earlier drafts.

9. Risk, Failure Recovery, Ethics, Security

Risk	Level	Mitigation
AAOIFI redistribution ambiguity	High	Publish manifests/hashes/scripts, not raw text, unless rights confirmed
No qualified Shari'ah validation	Critical	Require expert review before any headline claim
Small evaluation set	High	Treat pilot as exploratory; expand before publication
Long-context baseline beats RAG	High	Reframe as diagnostic/traceability contribution or negative result — not hidden
Conformal overclaim	High	Use careful selective-prediction language only
Model access/runtime issue	Moderate	Pin model path/revision and test access early
Citation-entailment false positives	Moderate	Human audit sample
Over-engineering	Moderate	Keep D+ retry and QLoRA strictly optional

Failure Recovery

- SAC underperforms → retain clause-level chunking for citability, report SAC as a negative result.
- UQ doesn't reduce residual error → report it as unreliable under AAOIFI retrieval conditions, pivot focus to retrieval/citation entailment.
- Disagreement triggers over-abstain → refine tags with expert input, report the coverage cost honestly.
- Long-context wins → reposition around evaluation, abstention behavior, cost/latency, and citation-entailment diagnostics rather than performance superiority.
- Licensing blocks release → publish code, synthetic examples, annotation schema, reproducibility instructions without protected text.

Ethics

The system must never issue a fatwa or final compliance ruling; outputs are labeled research assistance only; ambiguous/high-impact/disputed cases must escalate. Bias risks named explicitly: Sunni-centric benchmark coverage (inherited from IslamicMMLU), Arabic/English asymmetry, underrepresentation of minority jurisprudential positions. Scope is stated plainly rather than implying universal Islamic legal coverage.

Security

Retrieved standards are treated as data, not instructions (prompt-injection defense); all tool calls are logged; copyrighted source text is not exposed in public logs; deployment beyond research would require authentication, audit logs, role-based access, and legal/compliance review.

10. Dataset Strategy and Reproducibility

Source	Use	License / Risk
AAOIFI SS 8,9,13,17,26	Primary corpus	Public access; redistribution rights must be reviewed before publishing raw text
SAHM	Auxiliary QA/calibration	License must be re-checked at use time; prior review indicated non-commercial (CC BY-NC 4.0) constraints
Hand-built hard set	Main abstention/disagreement eval	Expert-labeled; publish questions/labels only if they don't reproduce protected text
IslamicLegalBench / FiqhQA / IslamicMMLU	Literature/topic guidance	Conceptual alignment only, not direct AAOIFI labels
Public prototypes (e.g. reported screening tools)	Prior-art context	Not peer-reviewed; cite as implementation prior art only, and only where independently corroborated (see Section 1)

Publication-grade minimum: $n \geq 100$ –150 expert-validated items; $n = 25$ –30 pilot allowed only for early prototyping, never for headline claims.

Reproducibility Package

- Environment file + package versions; model IDs, revisions, access notes; corpus manifest with hashes/dates; prompt templates + routing thresholds; random seeds; evaluation scripts; annotation guide; data card + model card; full logs for every final evaluation item. Repository excludes raw AAOIFI text unless redistribution rights are confirmed.

11. Publication Strategy and Academic Value

Near-term target: arXiv-style technical report + public GitHub artifact. **Realistic venue fit:** ArabicNLP, Natural Legal Language Processing, FinNLP-adjacent workshops, Islamic finance/AI or responsible-AI workshops, demo/shared-task tracks. **Poor near-term fit:** ACL/EMNLP/NeurIPS/ICML main tracks, unless the project grows into a larger benchmark or a genuinely new method. Never claim acceptance or submission before it happens — an important discipline this plan states explicitly.

Scholarship-application framing (the actual near-term goal): this project signals research maturity — realistic scoping, falsifiable design, honest handling of failure modes — continuous with the applicant's existing production AI/LLM background. A linkable, working GitHub repo with a real ablation table is more persuasive in an SOP than an unverifiable claim of “research in progress,” and should be described exactly as what it is: an open research prototype with a public technical note, not a publication.

12. Limitations

- Limited to five AAOIFI standards; does not validate actual financial products for real-world use.
 - Pilot hard set (n=25–30) is too small for strong statistical claims — directional/paired-test conclusions only.
 - Expert (qualified Shari'ah scholar) validation is mandatory before any authoritative domain claim, and is not guaranteed to be obtainable within the 4-week window.
 - A long-context baseline may reduce RAG's apparent value for a corpus this small — treated as a required test, not hidden.
 - SAHM and AAOIFI licensing may restrict redistribution of derived material.
 - Coverage cannot be claimed beyond the five in-scope standards, schools, or institutional interpretations actually evaluated.
 - Sunni-centric coverage is inherited wherever the plan draws on IslamicMMLU.
 - Roughly a fifth of the full reference list across both source documents was checked for existence/venue only, not for exact page-level precision — flagged plainly in Section 1 rather than smoothed over.
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13. Final Implementation Blueprint

- Begin with licensing and source-manifest design.
- Build retrieval evaluation before generation.
- Freeze a pilot hard set and annotation rubric before touching the generation layer.
- Implement single-pass answer-or-escalate routing first; add optional retry only once that core is stable.
- Add citation-entailment checks before any optional retry work.
- Run A, B, B', C, D, LC, and FRONTIER — the long-context and frontier rows are required, not aspirational.
- Treat all pilot findings as exploratory until the $n \geq 100$ –150 expert-labeled set exists.
- Publish code and derived metadata only within confirmed licensing limits.
- Submit to a workshop/demo venue only after a real, dated citation check like Section 1 — not after another round of self-declared "final" status.

14. Research Quality Score (Not Inflated)

Dimension	/10	Basis
Citation authenticity	9	13 of 14 independently checked items confirmed cleanly; 1 corrected error, 1 unconfirmable — this is the score, not 10.
Problem significance	6	Real, narrow market; not broadly high-impact beyond Islamic fintech compliance.
Novelty	5	Application/evaluation-level combinatorial novelty only, per Section 5.
Methodology	8	Required hostile baselines (long-context, frontier), staged n, paired testing — genuinely rigorous for the scale.
Experimental rigor	7	Correctly staged pilot-vs-main split; still capped by pilot sample size until Stage 2 executes.
Reproducibility	8	Strong: manifests, hashes, seeds, licensing discipline.
Feasibility (4-week core)	7	Realistic scope given free-tier compute and a single committed model.
Publication potential (top-tier)	3	Not realistic in current form.
Publication potential (workshop/shared-task)	7	Realistic and appropriate fit.
Overall as a research contribution	6	Up from a prior 5, mainly because required hostile baselines and staged evaluation close real methodological gaps identified earlier — not because of the citation re-check alone.
Overall as a scholarship-application artifact	8	Strong signal of research maturity, honest self-critique, and technical continuity with the applicant's existing profile.

No “final, green, 100% correct” status is issued here, and none should be sought from any future pass. The honest status is: core content is solid and has now survived two independent citation checks with one real error found and fixed; a small number of citations remain checked-for-existence-only rather than checked-for-precision; two named risks (expert validation, small pilot n) are structural and cannot be resolved by more editing — only by actually doing Stage 2 of the plan.

15. References (With Verification Status)

Abdelaal, A., Al Haffar, M. N., Fawzi, M., & Magdy, W. (2026). IslamicMMLU: A Benchmark for Evaluating LLMs on Islamic Knowledge. arXiv:2603.23750.

Verification: *CONFIRMED — this audit, full text*

Al-Smadi, M. (2025). QU-NLP at QIAS 2025 Shared Task. arXiv:2508.15854.

Verification: *CONFIRMED — headline figures, this audit*

AAOIFI. Shari'ah Standards SS 8, 9, 13, 17, 26. aaoifi.com/e-standards.

Verification: *CONFIRMED — titles cross-checked via regulator and AAOIFI documents, this audit*

Atif, F., Askarbekuly, N., Darwish, K., & Choudhury, M. (2025). Sacred or Synthetic? (FiqhQA). AIES 2025, 8(1), 217–226. arXiv:2508.08287.

Verification: *CONFIRMED — this audit, full text*

Anwar, M. et al. (2026). Jais 2: A Family of Arabic-Centric Open Large Language Models. arXiv:2608.13580.

Verification: *CONFIRMED — this audit, including inception42 org name*

Bhatia, G., Mubarak, H., Jarrar, M., et al. (2026). From RAG to Agentic RAG for Faithful Islamic QA. Findings of ACL 2026. arXiv:2601.07528.

Verification: *CONFIRMED — full text, exact figures; exact page numbers within volume not re-pinned*

Elbadry, R., et al. (2026). SAHM: A Benchmark for Arabic Financial and Shari'ah-Compliant Reasoning. ACL 2026 (2026.acl-long.1593). arXiv:2604.19098.

Verification: *CONFIRMED — this audit, incl. ACL Anthology acceptance*

Elmahjub, E., Qadir, J., Mushtaq, A., et al. (2026). IslamicLegalBench. arXiv:2602.21226.

Verification: *CONFIRMED — this audit, full text*

Es, S., et al. (2024). RAGAs: Automated Evaluation of RAG. EACL 2024 System Demonstrations, 150–158.

Verification: *CONFIRMED (existence/venue) — not re-derived page-by-page this pass*

Nguyen, V., Dang, C., Zhang, J., et al. (2026). URAG. arXiv:2603.19281.

Verification: *CONFIRMED — this audit, full text incl. all 9 authors*

Reuter, M., Lingenberg, T., Liepiņa, R., et al. (2025). Towards Reliable Retrieval in RAG Systems for Large Legal Datasets (SAC). NLLP 2025. arXiv:2510.06999.

Verification: *CONFIRMED — this audit*

Tayebati, S., Kumar, D., Darabi, N., et al. (2026). CAP. PMLR 304, 926–941.

Verification: *CONFIRMED (existence/venue) — not re-derived page-by-page this pass*

Vashurin, R., Fadeeva, E., Vazhentsev, A., et al. (2025). LM-Polygraph. TACL, 13, 220–248.

Verification: *CONFIRMED (existence/venue) — seen consistently across multiple independent citing sources*

Vashurin, R., Goloburda, M., Ilina, A., et al. (2025). CoCoA: A Minimum Bayes Risk Framework Bridging Confidence and Consistency for UQ in LLMs. **NeurIPS 2025** (corrected from "NeurIPS 2026" in a prior draft).

Verification: *CONFIRMED WITH CORRECTION — this audit, via official NeurIPS/OpenReview listing*

Stankovic, M. (2026). Cross-Document Topic-Aligned Chunking for Retrieval-Augmented Generation. arXiv:2601.05265.

Verification: *CONFIRMED (existence) — optional context only, non-blocking*

Reported implementation prior art (rules-as-code Shari'ah screening; commercial screening tools).

Verification: *NOT INDEPENDENTLY VERIFIABLE — could not corroborate specific repository claims this session; treat as unconfirmed context, not a citation*